

Question	Answer	Marks
1(a)	a scalar has magnitude (only)	B1
	a vector has magnitude and direction	B1
1(b)	power: scalar temperature: scalar momentum: vector <i>(two correct 1 mark, all three correct 2 marks)</i>	B2
1(c)(i)	arrow labelled R in a direction from 5° to 20° north of west	B1
1(c)(ii)	$v^2 = 28^2 + 95^2 - (2 \times 28 \times 95 \times \cos 115^\circ)$ or $v^2 = [(95 + 28 \cos 65^\circ)^2 + (28 \sin 65^\circ)^2]$	C1
	$v = 110 \text{ ms}^{-1}$ (109.8 ms ⁻¹)	A1
	or (scale diagram method)	
	triangle of velocities drawn	(C1)
	$v = 110 \text{ ms}^{-1}$ (allow 108–112 ms ⁻¹)	(A1)

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Question	Answer	Marks
2(c)	a body continues at (constant) velocity unless acted upon by a resultant force	D1